CITS4404 Project — Building AI Trading Bots: Nature-Inspired Optimisation Algorithms Implementation and Experimentation, and Evaluation

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2421 words

I. ALGORITHMS INVESTIGATED

Six optimisation algorithms were evaluated across all experimental configurations, spanning four structural categories: swarm-based, physics-inspired, ecosystem-inspired, and classical local search.

Particle Swarm Optimization, originally proposed by Kennedy and Eberhart [1] and extended with an inertia weight by Shi and Eberhart [2], is a swarm method in which each particle maintains both a personal best position and a shared global best. The inertia weight w governs the balance between exploration and exploitation through the velocity update rule. Atomic Orbital Search (AOS) [3] is a physics-inspired method detailed in Section III; it employs shell-based clustering around a nucleus (global best) as its primary exploration mechanism, without per-agent personal memory. African Buffalo Optimization (ABO) [4] encodes two herd vocalisations — an exploratory foraging call and an exploitative regrouping call — and like PSO retains both personal and global best positions. Manta Ray Foraging Optimization (MRFO) [5] models three foraging behaviours (chain, cyclone, and somersault foraging) but maintains only a global best, with no per-agent memory. Symbiotic Organisms Search (SOS) [6] simulates mutualistic, commensal, and parasitic ecological interactions, applying all three phases to every organism per iteration; this yields three fitness evaluations per organism per iteration and requires no algorithm-specific hyperparameters. Generalized Pattern Search (GPS) [7] is a classical single-state local optimiser that evaluates a fixed pattern of directions from the current best solution, with no population and no historical memory beyond the current position; it serves as a deliberate baseline to quantify the value of population-based diversity.

Table I summarises the distinguishing characteristics of each algorithm.

TABLE I. Summary of the six optimisation algorithms evaluated. Memory column describes which historical positions are retained per agent. Eval/iter denotes fitness evaluations per agent per iteration.

Algorithm	Year	Category	Population-based	Memory	Eval/iter
AOS	2021	Physics	Yes	Global (nucleus)	1
PSO	1998	Swarm	Yes	Personal + Global	1
MRFO	2020	Swarm	Yes	Global only	1
SOS	2014	Ecosystem	Yes	Global only	3
ABO	2015	Swarm	Yes	Personal + Global	1
GPS	—	Local search	No	Current best only	1

II. BOT DESIGN AND PARAMETRISATION

A. Trading Signal and Fitness

Each strategy produces a short-window (low-period) and a long-window (high-period) smoothed price series. The trading signal detects crossover events in their difference using a two-point weighted kernel:

$$d_t = \text{Series}_{\text{short},t} - \text{Series}_{\text{long},t} \quad , \quad C_t = \tfrac{1}{2}(\text{sign}(d_t) - \text{sign}(d_{t-1})) \tag{1}$$

$$S = \begin{cases} \text{buy} & \text{if } C_t = -1 \text{ (short crosses below long)} \\ \text{sell} & \text{if } C_t = +1 \text{ (short crosses above long)} \end{cases} \tag{2}$$

where d_t is the difference of the two smoothed series at time t, $C_t \in \{-1, 0, +1\}$ is the crossover indicator, and S is the resulting trade action. The signal is non-zero only at the moment of a crossover; it is zero on all other timesteps.

The fitness function measures final USD holdings after simulating trades over the full training price series:

$$\text{Fitness}(X) = \text{USD}_{\text{final}} \tag{3}$$

where X is the candidate parameter vector, $\text{USD}_{\text{final}}$ is the cash held at the end of the simulation, and the starting balance is $\text{USD}_{\text{initial}} = \$1000$. A transaction fee of 3% is applied to each trade. The simulation is a single continuous pass over the training data with no mid-run resets.

One acknowledged limitation is that the fitness function imposes no penalty for non-trading: if no crossover ever occurs, $C_t = 0$ for all t and the bot holds cash throughout, returning $\text{Fitness}(X) = \text{USD}_{\text{initial}}$. This is a valid but uninformative fixed point. The bad/broken optimum was observed empirically — MRFO converged to parameter regions with no crossover signal under certain configurations. A risk-adjusted metric such as the Sharpe ratio would penalise this behaviour but was not adopted here; the limitation is documented rather than silently corrected.

B. Strategy Configurations

Five strategy configurations define the hypothesis space presented to each optimiser.

Strategy 1 — Double SMA crossover ($d = 2$). Both the short window $x_0 \in [2, 50]$ and the long window $x_1 \in [51, 200]$ [8] are optimised. The non-overlapping constraint (short < long) is guaranteed by construction through the parameter bounds, with no runtime enforcement required. The SMA assigns uniform weight to all prices within the window, making it insensitive to the ordering of recent versus older prices.

Strategy 2 — Double LMA crossover ($d = 2$). The parameter structure is identical to Strategy 1 ($x_0 \in [2, 50]$, $x_1 \in [51, 200]$). The LMA assigns linearly decaying weights so that more recent prices contribute more than older ones, providing momentum sensitivity that the uniform SMA cannot capture. The optimiser thus faces the same two-dimensional search space as Strategy 1 but with a smoother series that responds differently to regime changes.

Strategy 3 — EMA with shared α ($d = 3$). A short span, a long span, and a single shared smoothing parameter $\alpha \in (0, 1)$ are optimised jointly. Sharing one α value across both EMAs constrains both series to decay at identical rates; the frequency contrast that drives the crossover signal therefore depends entirely on the span difference rather than on independent decay dynamics. This effectively collapses the degree of freedom to approximately 2.5 effective dimensions in practice.

Strategy 4 — EMA with independent α ($d = 4$). Short span, long span, α_{short}, and α_{long} are each optimised independently, along with additional EMA window parameters. Decoupling the smoothing parameters gives the short-term EMA its own reactivity and the long-term EMA its own stability, restoring the frequency contrast that shared α eliminates. The expanded search space ($d = 4$ versus $d = 3$) carries a higher exploration cost but provides genuinely greater expressiveness: the two EMAs can now operate at qualitatively different timescales simultaneously. Empirically, this configuration produced the highest Kruskal-Wallis H-statistics across all strategy conditions, suggesting that the larger search space amplifies inter-algorithm performance differences rather than masking them.

Strategy 5 — Weighted MA combination ($d = 14$). SMA, LMA, and EMA signals are combined as a weighted sum with learned weights w_1, w_2, w_3. All three weights are treated as free optimisation dimensions and normalised at evaluation time (each w_i divided by $w_1 + w_2 + w_3$), yielding a 14-dimensional parameter vector that includes window parameters for each MA type plus EMA spans and smoothing factors. Deriving $w_3 = 1 - w_1 - w_2$, as the project brief implies, would create asymmetric optimisation pressure because w_3 is never directly searched; retaining all three as free variables ensures each weight receives equal search effort. The 14-dimensional space is the largest tested, and the curse of dimensionality is a genuine concern when each optimiser is limited to 50 runs.

C. Justified Deviations from Project Brief

TABLE II. Summary of implementation decisions that deviate from the project brief, with justification for each.

Deviation	Brief approach	Our implementation	Justification
EMA normalisation	Optional	Enabled by default	Unnormalised kernels produce span-dependent amplitude distortion, making fitness values non-comparable across parameter configurations
$d = 14$ weight derivation	w_3 derived as $1 - w_1 - w_2$	All three weights free; normalised at eval time	Derived w_3 receives no direct search pressure, creating asymmetric optimisation bias
n_{shells} assignment	Fixed shell count	$\min(n_{\text{shells}}, \lvert\text{remaining}\rvert)$	Prevents degenerate empty shell states in high-dimensional runs where population may be smaller than the configured shell count
EMA α across spans	Shared α (Strategy 3)	Independent α_{short}, α_{long} (Strategy 4)	Shared α constrains both series to identical decay rates, eliminating the frequency contrast that makes the crossover signal informative

III. ALGORITHM SELECTION

All six algorithms described in Section I were evaluated. This section explains the rationale behind each inclusion and details the primary algorithm, AOS.

GPS [7] as a deliberate baseline.: with no population, no escape mechanism, and no positional memory beyond the current best, it directly quantifies whether population-based diversity is necessary on this landscape.

*PSO [2] as a population-based benchmark.*and a structurally informative paired comparison with AOS as both use a global best to guide the population, so any performance difference isolates the contribution of per-particle personal-best memory.

AOS as the physics-based algorithm compared to the others being biology-based. The MA parameter fitness landscape is expected to be multimodal: distinct parameter combinations can produce similar short-term profits through qualitatively different mechanisms, creating multiple local peaks of comparable height. AOS [3] was selected because its shell-based population structure explicitly supports simultaneous exploitation of multiple landscape regions. Candidate solutions (electrons) are assigned to concentric shells around the nucleus (current global best) according to their fitness rank; electrons in inner shells exploit promising regions while outer-shell electrons conduct broader exploration. Transitions between shells are governed by a quantum-mechanical photon absorption and emission analogy: an electron absorbs energy (moves to an outer shell, exploration) when its transition energy falls below the shell's binding energy, and emits energy (moves to an inner shell, exploitation) otherwise. This structured push-pull between shells provides more persistent population spread than PSO's convergent swarm behaviour, which collapses toward the global best over time. AOS was published in 2021 and has not, to our knowledge, been applied to financial parameter optimisation in the literature.

ABO, MRFO, and SOS are detailed in Deliverable 1 their characteristics are summarized in Table I

The pseudocode below summarises the core AOS iteration:

```
Initialise N electrons randomly in the search space
Set nucleus := position of the best electron (highest fitness)
Assign electrons to K shells by fitness rank (binding energy)

FOR each iteration t = 1, 2, ..., T:
    FOR each shell k = 1, ..., K:
        Compute shell binding energy E_k
        FOR each electron i in shell k:
            Sample transition energy E
            IF E < E_k  (electron absorbs energy):
                Move to outer shell; apply exploration step
            ELSE  (electron emits energy):
                Move to inner shell; apply exploitation step
            Update position via quantum transition equation
            Evaluate fitness of new position
    Apply LE perturbation: randomly displace the
        lowest-energy electron to escape local optima
```

```
    Update nucleus if any electron improves best fitness

RETURN nucleus position as best solution found
```

LE degradation — an observed weakness. The Lowest Energy (LE) perturbation step is designed to prevent stagnation by randomly displacing the electron with the worst fitness. In practice, the LE designation is not stable: as other electrons improve their fitness across iterations, the electron currently labelled LE may no longer occupy the position most in need of escape relative to the updated nucleus. The original paper [3] does not address this drift. This was observed empirically across runs and is logged per iteration; it is documented here rather than patched, as any correction would constitute a modification to the published algorithm.

Metaparameter choices. All algorithms were run with a population of 100 and a fixed iteration budget, ensuring a controlled per-iteration comparison. The shell count n_{shells} is implemented as an upper cap rather than a fixed assignment: each iteration uses $\min(n_{\text{shells}}, |\text{remaining population}|)$ shells to prevent degenerate empty-shell states that can arise in high-dimensional runs where effective population diversity is reduced.

IV. EXPERIMENTS AND EVALUATION

A. Dataset

Daily BTC/USD OHLCV data (Kaggle, 2014-2022) was used, with training on pre-2020 data and testing on 2020-2022. The post-2020 period encompasses three qualitatively distinct regimes — the COVID-19 crash, the 2021 bull run, and the 2022 bear market — constituting an intentional out-of-distribution stress test rather than a random holdout.

B. Experimental Design

The full factorial design comprised 6 algorithms $\times$ 5 strategies $\times$ 2 initialisation modes $\times$ 50 runs = 3000 total trials. Fixed initialisation seeds all runs from a saved initialized once position; random initialisation draws each run uniformly from the search space. Strong dependence on initialisation mode would indicate algorithmic fragility. A run count of 50 balances statistical power against computational cost.

C. Statistical Analysis

Kruskal-Wallis was used in preference to ANOVA because optimiser output systematically violates normality: runs can converge to the no-trade fixed point ($1000), produce occasional high-value jackpot outliers, or spread heavily in between — all genuine behavioural modes, not noise. Kruskal-Wallis operates on ranks and is robust to these distributional shapes. The test was applied per strategy $\times$ initialisation combination with algorithm as the factor, isolating algorithmic effects from strategy-level confounds. Pairwise post-hoc comparisons used Dunn's test with Bonferroni correction ($\alpha = 0.05$).

V. RESULTS AND ANALYSIS

A. Training Results

The Kruskal–Wallis test was significant across all ten strategy $\times$ initialisation combinations on training fitness ($p < 10^{-34}$ in every case; Table III). Three patterns emerge from Dunn's pairwise post-hoc comparisons (Figure 1). First, GPS is significantly separated from all population-based methods in the large majority of conditions. The clearest exception is `ema_independent`, where GPS and MRFO are not significantly different ($p = 0.16$ fixed, $p = 1.00$ random); MRFO collapses to the no-trade fixed point in every fixed-initialisation run (CV = 0), so the non-significance reflects two algorithms failing to the same floor.

Second, the PSO–SOS–AOS cluster is strategy-dependent: largely indistinguishable on `weighted` and $d = 2$ strategies, but breaking apart on `ema_independent`, where AOS separates significantly from PSO ($p = 1.7 \times 10^{-3}$ fixed, $p = 2.5 \times 10^{-6}$ random) and from SOS.

FIG. 1. Dunn's post-hoc pairwise p-values (Bonferroni-corrected) for all strategy × initialisation combinations — training fitness. Red indicates significant difference ($p < 0.05$); green indicates no significant difference.

Third, SOS exhibits zero within-cell variance on $d = 2$ strategies (CV = 0.000; Table IV), consistent with its three-phase update collapsing diversity rapidly. Its mean wall-clock runtime (14,730 ms) is roughly four times that of PSO (3,706 ms).

B. Test Results — Performance Against the Passive Baseline

A passive buy-and-hold strategy over the 2020–2022 test period produced \$5,660.26 from a \$1,000 starting balance. Of the 3,000 optimised bots, only 62 (2.07%) strictly exceeded this baseline; no algorithm × strategy median exceeded it under any condition (Figure 2). Win rates were heavily stratified by strategy (Table V): `sma` and `lma` yielded zero wins across 1,200 combined runs; `ema_independent` produced only 2/600; `weighted` produced 26/600; and `ema_shared` produced 34/600.

TABLE III. Kruskal–Wallis H-test results comparing optimisation algorithms for each strategy and initialisation mode, on *training* and *test* fitness. On training fitness, algorithm choice is universally significant ($p < 10^{-34}$). On test fitness the effect is strategy-dependent: `ema_shared` loses all significance ($p > 0.90$), while `ema_independent` retains the strongest effect ($H > 179$).

		Training		**Test**	
Strategy	**Start**	H	p	H	p
sma	fixed	200.63	2.1×10^{-41}	46.59	6.9×10^{-9}
sma	random	198.09	7.3×10^{-41}	85.68	5.4×10^{-17}
lma	fixed	176.56	2.9×10^{-36}	73.95	1.5×10^{-14}
lma	random	170.28	6.3×10^{-35}	58.34	2.7×10^{-11}
ema_shared	fixed	206.76	1.0×10^{-42}	0.60	9.9×10^{-1}
ema_shared	random	190.41	3.2×10^{-39}	1.51	9.1×10^{-1}
ema_independent	fixed	223.17	3.1×10^{-46}	184.26	6.6×10^{-38}
ema_independent	random	244.91	6.8×10^{-51}	179.13	8.2×10^{-37}
weighted	fixed	213.19	4.3×10^{-44}	78.91	1.4×10^{-15}
weighted	random	224.80	1.4×10^{-46}	78.54	1.7×10^{-15}

C. Test Results — Statistical Differentiation

On test data, the strong training differences largely collapse (Figure 4; Table III). The most extreme case is `ema_shared`: every pairwise p-value equals 1.000 and Kruskal–Wallis is non-significant under both initialisation modes — all six algorithms are statistically indistinguishable on the strategy that produced the most baseline-beating runs.

The one strategy retaining strong test-side differentiation is `ema_independent` ($H = 184.26$ fixed, 179.13 random). PSO and SOS retain high test medians (pooled medians \$4,435 and \$4,566), whereas AOS collapses to \$1,105 despite leading on training fitness (Table VI). A caution applies: on strategies where roughly 32% of runs sit at broken fixed points, statistical significance can reflect how often an algorithm collapses rather than genuine optimisation quality.

D. Train–Test Rank Transfer

Spearman rank correlation (Figure 5) reveals that `sma` has the strongest train–test transfer ($\rho = 0.66$), while `ema_shared` shows none ($\rho = 0.05$, $p = 0.20$, n.s.). The relationship is not monotonic in dimensionality: `ema_shared` ($d = 3$) transfers worse than the $d = 14$ `weighted` space ($\rho = 0.34$), and decoupling the two EMAs' smoothing rates moves ρ from 0.05 to 0.38 with no other change. The nature of the parameterisation matters more than its size.

Notably, win rate is anti-correlated with transfer reliability: the two strategies with the most baseline-beating runs (`ema_shared`, `weighted`) have the weakest transfer, while `sma` — the strongest transfer — produced zero wins. Table VII makes this concrete: the single highest test result (\$9,615, GPS `ema_shared`) came from a training fitness of only \$16,290 — an order of magnitude below runs that tested far lower.

TABLE IV. Summary of *training* fitness by algorithm and strategy, pooled across both initialisation modes (100 runs per cell). Note the very high training fitness reached on `ema_shared` (medians up to $133,000), which Section V shows does not transfer to test.

Algo	Strategy	Mean	Std	CV	Median	Max
abo	ema_independent	10,750.12	1,802.05	0.168	10,347.23	15,352.63
aos	ema_independent	19,631.96	5,008.31	0.255	21,904.21	21,904.21
gps	ema_independent	5,179.20	3,858.59	0.745	4,543.98	12,939.69
mrfo	ema_independent	1,000.00	0.00	0.000	1,000.00	1,000.00
pso	ema_independent	12,092.59	1,255.72	0.104	11,463.70	15,303.77
sos	ema_independent	13,273.95	799.34	0.060	13,197.07	15,607.47
abo	ema_shared	93,661.07	24,627.52	0.263	90,428.45	157,746.47
aos	ema_shared	67,128.07	26,684.54	0.398	60,697.82	229,103.39
gps	ema_shared	11,989.89	14,016.59	1.169	7,882.87	59,090.09
mrfo	ema_shared	86,615.44	31,961.19	0.369	78,405.64	237,614.46
pso	ema_shared	103,646.40	22,819.48	0.220	102,724.96	193,483.44
sos	ema_shared	132,840.58	30,893.95	0.233	128,243.17	215,646.97
abo	lma	802.79	16.63	0.021	800.11	818.93
aos	lma	805.44	15.38	0.019	809.92	818.93
gps	lma	463.95	120.20	0.259	461.38	818.93
mrfo	lma	810.79	11.45	0.014	818.93	818.93
pso	lma	798.10	16.98	0.021	800.11	818.93
sos	lma	818.93	0.00	0.000	818.93	818.93
abo	sma	930.41	13.21	0.014	932.50	932.50
aos	sma	931.76	0.66	0.001	931.18	932.50
gps	sma	551.42	118.34	0.215	542.81	917.97
mrfo	sma	932.50	0.00	0.000	932.50	932.50
pso	sma	909.75	48.31	0.053	932.50	932.50
sos	sma	932.50	0.00	0.000	932.50	932.50
abo	weighted	11,339.38	1,200.41	0.106	11,385.64	14,205.09
aos	weighted	10,697.16	2,147.99	0.201	10,355.90	29,952.87
gps	weighted	5,124.90	3,544.28	0.692	5,295.31	12,601.00
mrfo	weighted	14,049.10	9,313.82	0.663	11,952.69	65,347.85
pso	weighted	12,744.28	1,112.69	0.087	12,602.90	17,561.96
sos	weighted	13,215.13	549.20	0.042	13,242.64	14,531.01

TABLE V. Runs exceeding the passive buy-and-hold baseline under strict > comparison. The baseline is $5,660.26; counts are out of 50 runs per cell; 3,000 trials in total.

	ema_ind		**ema_shared**		**lma**		**sma**		**weighted**	
algo	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix	rnd
abo	0	0	1	3	0	0	0	0	3	6
aos	0	0	1	3	0	0	0	0	2	2
gps	2	0	3	3	0	0	0	0	0	0
mrfo	0	0	3	3	0	0	0	0	0	3
pso	0	0	2	6	0	0	0	0	2	1
sos	0	0	4	2	0	0	0	0	4	3
total	**2**	**0**	**14**	**20**	**0**	**0**	**0**	**0**	**11**	**15**

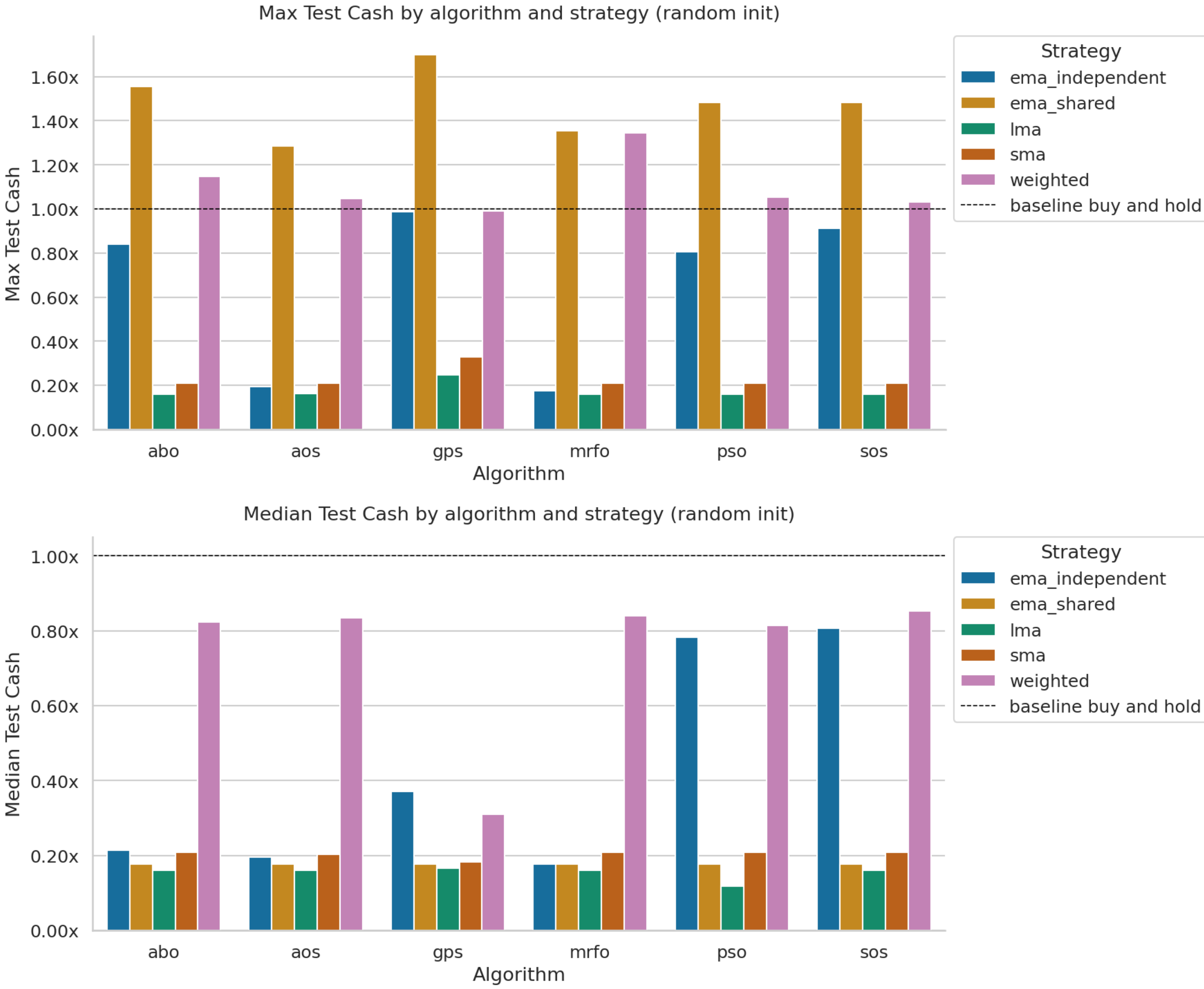

FIG. 2. Median test cash by algorithm and strategy, normalised to the buy-and-hold baseline (\$5,660.26 from \$1,000 initial under matched 3% entry and exit fees), for fixed (top) and random (bottom) initialisation. No algorithm × strategy median exceeds the baseline (1.00× line) under any condition.

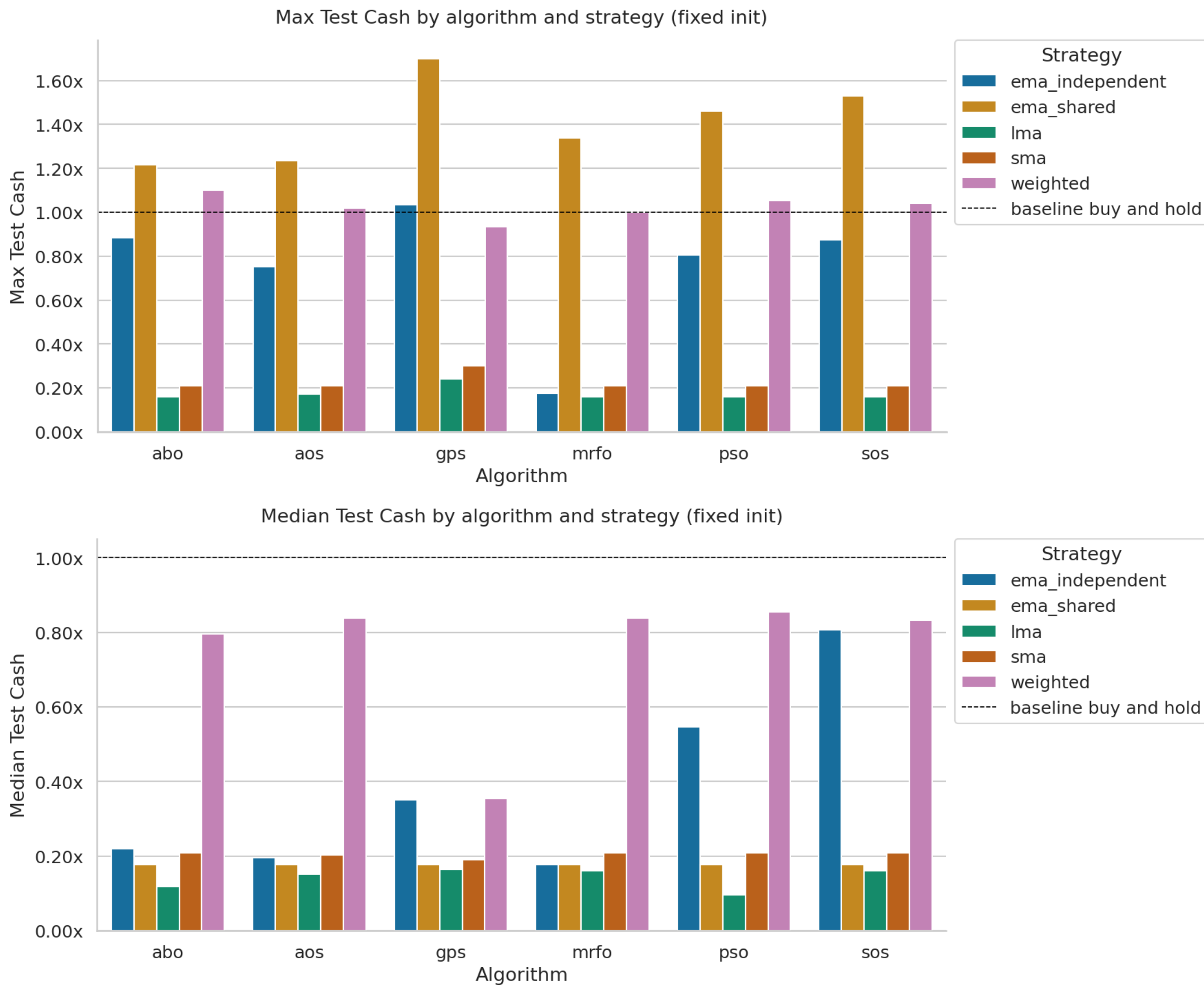

FIG. 3. Maximum test cash by algorithm and strategy, normalised to the buy-and-hold baseline, for fixed (top) and random (bottom) initialisation. Bars above 1.00× indicate at least one run in 50 exceeded passive holding; concentration of such bars in `ema_shared` reflects its long-tailed jackpot distribution rather than reliable optimisation.

TABLE VI. Summary of *test* fitness by algorithm and strategy, pooled across both initialisation modes (100 runs per cell). Compare against Table IV: training medians above $100,000 on `ema_shared` collapse to a test median of $1,000 for every algorithm.

Algo	Strategy	Mean	Std	CV	Median	Max
abo	ema_independent	1,455.93	800.27	0.550	1,239.42	4,998.46
aos	ema_independent	1,079.86	496.95	0.460	1,104.87	4,249.85
gps	ema_independent	2,633.91	1,703.55	0.647	1,982.11	5,850.78
mrfo	ema_independent	1,000.00	0.00	0.000	1,000.00	1,000.00
pso	ema_independent	3,288.24	1,485.39	0.452	4,435.27	4,565.80
sos	ema_independent	3,806.42	1,374.95	0.361	4,565.80	5,159.96
abo	ema_shared	1,903.99	1,636.50	0.860	1,000.00	8,795.49
aos	ema_shared	1,914.91	1,653.88	0.864	1,000.00	7,284.67
gps	ema_shared	1,970.13	2,187.39	1.110	1,000.00	9,614.94
mrfo	ema_shared	1,868.67	1,771.49	0.948	1,000.00	7,660.07
pso	ema_shared	2,265.04	1,993.23	0.880	1,000.00	8,392.40
sos	ema_shared	2,190.03	2,061.31	0.941	1,000.00	8,663.88
abo	lma	718.72	186.57	0.260	671.51	907.98
aos	lma	799.00	173.74	0.217	907.98	977.77
gps	lma	906.20	217.94	0.241	931.86	1,399.39
mrfo	lma	759.18	197.79	0.261	907.98	907.98
pso	lma	659.46	174.85	0.265	671.51	907.98
sos	lma	907.98	0.00	0.000	907.98	907.98
abo	sma	1,170.00	22.46	0.019	1,180.28	1,180.28
aos	sma	1,165.29	13.35	0.011	1,153.51	1,180.28
gps	sma	1,080.49	248.38	0.230	1,055.30	1,857.14
mrfo	sma	1,180.28	0.00	0.000	1,180.28	1,180.28
pso	sma	1,112.79	155.32	0.140	1,180.28	1,180.28
sos	sma	1,180.28	0.00	0.000	1,180.28	1,180.28
abo	weighted	4,453.79	1,124.67	0.253	4,620.62	6,498.67
aos	weighted	4,606.32	831.97	0.181	4,745.43	5,920.21
gps	weighted	2,302.15	1,651.62	0.717	1,891.33	5,610.75
mrfo	weighted	4,570.30	977.47	0.214	4,755.59	7,604.38
pso	weighted	4,510.11	993.36	0.220	4,766.00	5,965.44
sos	weighted	4,840.99	516.32	0.107	4,749.80	5,889.75

TABLE VII. The 20 highest test-fitness runs across all 3,000 trials. Training fitness is shown alongside for comparison: the ordering by test fitness bears little relation to the ordering by training fitness.

Algo	Strategy	Start	Train Cash	Test Cash
gps	ema_shared	fixed	16,290.48	9,614.94
gps	ema_shared	fixed	16,290.48	9,614.94
gps	ema_shared	random	16,290.48	9,614.94
gps	ema_shared	random	16,290.48	9,614.94
abo	ema_shared	random	54,301.41	8,795.49
sos	ema_shared	fixed	166,428.24	8,663.88
sos	ema_shared	fixed	128,331.33	8,392.40
sos	ema_shared	fixed	114,909.60	8,392.40
gps	ema_shared	fixed	47,972.89	8,392.40
pso	ema_shared	random	138,076.13	8,392.40
sos	ema_shared	random	87,164.40	8,392.40
gps	ema_shared	random	30,519.05	8,392.40
pso	ema_shared	fixed	80,647.43	8,267.02
mrfo	ema_shared	random	125,386.17	7,660.07
mrfo	weighted	random	48,030.39	7,604.38
mrfo	ema_shared	fixed	105,631.68	7,586.17
abo	ema_shared	random	71,590.32	7,499.73
aos	ema_shared	random	41,659.30	7,284.67
pso	ema_shared	random	115,538.41	7,226.59
pso	ema_shared	fixed	107,480.63	7,185.07

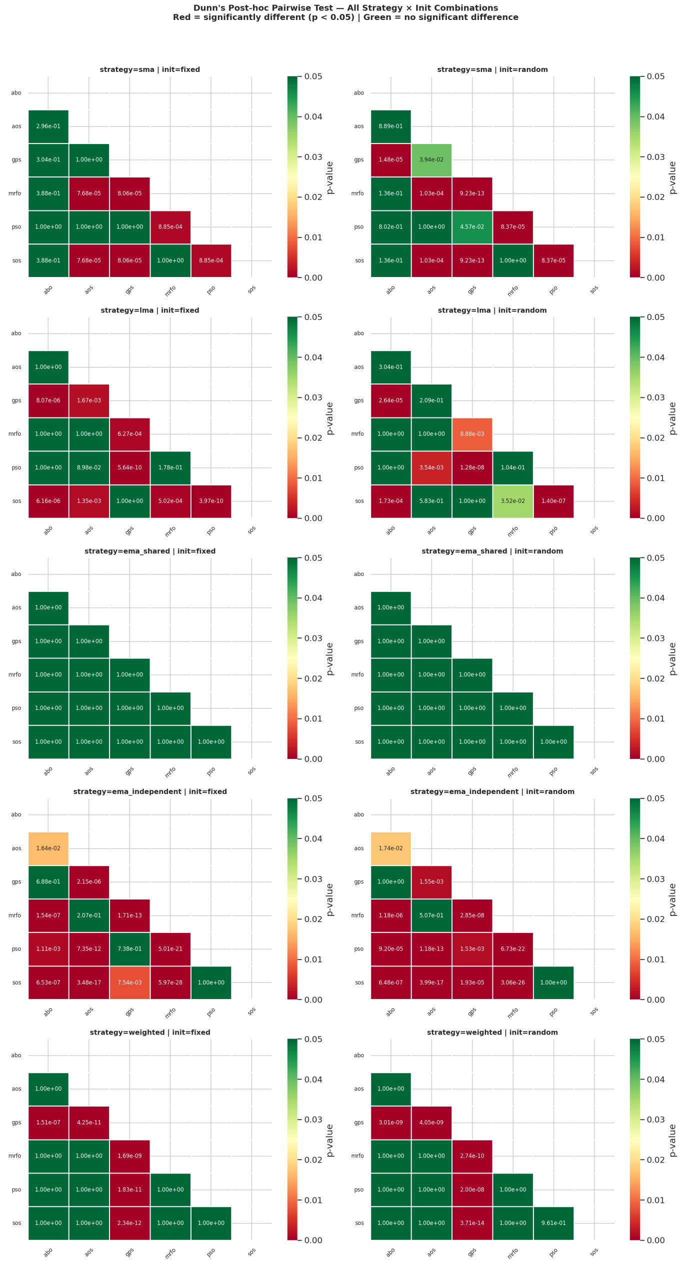

FIG. 4. Dunn's post-hoc pairwise p-values (Bonferroni-corrected) for all strategy $\times$ initialisation combinations — test fitness.

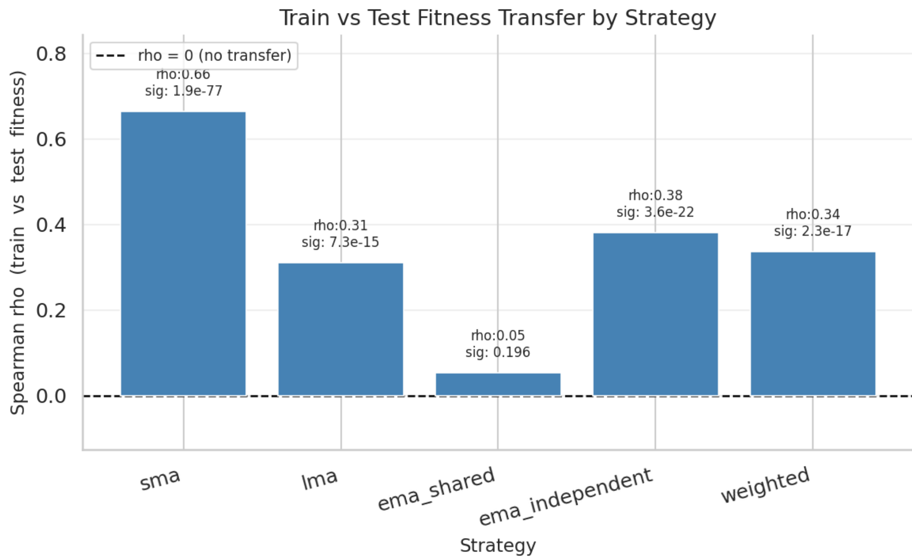

FIG. 5. Spearman rank correlation between training and test fitness per strategy (both initialisation modes pooled, $n = 600$ per strategy). $\rho = 0$ corresponds to no rank transfer.

VI. CONCLUSION

Algorithm choice has a universally significant effect on training fitness across all strategy and initialisation conditions, yet this advantage does not transfer to test performance. The binding constraint is strategy expressiveness in the face of market non-stationarity: once the target regime shifts, optimised parameters no longer generalise regardless of how effectively they were found.

The near-universal training-to-test failure is best attributed to a misspecified objective rather than defective optimisers. The decisive evidence is that the train-to-test gap scales with optimisation *success*, not algorithm identity: GPS, with the lowest training fitness, has a train/test ratio near 1.0; SOS, with the highest, shows the largest gap. Within `ema_shared`, training medians span \$7,900–\$128,000 while every algorithm's test median sits at \$1,000 (Tables IV, VI). Two non-exclusive mechanisms underlie this: overfitting to the 2014–2019 price path, and MA-crossover signals being fundamentally weak predictors across regimes. The 2.07% baseline-beating rate is consistent with right-tail sampling rather than a learnable edge — consistent with, though not a formal test of, the efficient market hypothesis for short-horizon technical signals.

AOS vs PSO. AOS performs comparably to PSO on training fitness despite lacking personal-best memory, suggesting that shell-based population distribution is sufficient on the lower-dimensional strategies. However, on `ema_independent` AOS collapses on test (median \$1,105) while PSO and SOS retain high medians (\$4,435 and \$4,566). Since the primary structural difference is per-agent memory, this suggests personal-best history aids generalisation on the most expressive landscape — though the present design does not isolate memory as a variable.

GPS as baseline. GPS is consistently and significantly worse than every population-based method on training fitness, confirming that population diversity is a structural requirement on this multimodal landscape, not an optional improvement. Its small train-to-test gap reflects weak optimisation on both splits rather than robustness.

SOS compute cost. SOS's three-evaluation-per-iteration overhead (wall-clock $\approx 4\times$ PSO) is not justified by its performance: Dunn's test finds no significant advantage over PSO under matched-iteration budgets.

Parameterisation structure. Independent α in `ema_independent` provides greater expressiveness than shared α, producing higher H-statistics and cleaner algorithm separation. Critically, `ema_shared` ($d = 3$) transfers worse than the $d = 14$ `weighted` space ($\rho = 0.05$ vs 0.34): parameterisation *nature* matters more than size.

AOS-specific weakness. The LE perturbation degrades across iterations as the LE designation tracks population rank rather than an absolute landscape property, reducing its escape effectiveness as the nucleus shifts. This is a genuine weakness in the published formulation.

No Free Lunch. Strategy choice accounts for far more outcome variance than algorithm choice, consistent with the No Free Lunch theorem [9]: no algorithm dominates uniformly across landscapes.

Outlook Future work should adopt risk-adjusted fitness functions (e.g. Sharpe ratio [10]) and walk-forward or cross-validated evaluation to impose generalisation pressure. Separating the two failure mechanisms — objective misspecification vs. market non-stationarity — requires running the same optimisers against a risk-adjusted objective under the same split: if the gap narrows, the fitness function dominates; if it persists, regime shift dominates.

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